

Answers

Q	Ans	Q	Ans	Q	Ans
1	b	2	b	3	a
4	b	5	a	6	b
7	b	8	b	9	a
10	b	11	a	12	b
13	b	14	b	15	c
16	a	17	d	18	a
19	a	20	b	21	a
22	b	23	a	24	b
25	c	26	b	27	d
28	a	29	b	30	a
31	a	32	b	33	b
34	c	35	b	36	a
37	c	38	a	39	d
40	a	41	c	42	b
43	c	44	b	45	a
46	b	47	c	48	b
49	a	50	c	51	a
52	b	53	d	54	a
55	c	56	a	57	b
58	b	59	a	60	b

Selected working

(5) Range: $x \in \{1, 2, 3\} \rightarrow y = x + 1 \in \{2, 3, 4\}$.

(14) $f(4) = 3(4) - 2 = 10$.

(19) $(fg)(x) = (4x - 3)(2x + 5) = 8x^2 + 20x - 6x - 15 = 8x^2 + 14x - 15$.

(21) $(f \circ g)(x) = f(g(x)) = 2(x + 3) = 2x + 6$.

(22) $(g \circ f)(x) = g(f(x)) = (2x) + 3 = 2x + 3$.

(27) x^2 is not one-to-one over $\mathbb{R}$ ($(-1)^2 = 1^2$) and not onto (no $y < 0$).

(28) $2x + 3$ is a linear bijection over $\mathbb{R}$.

(30) $y = 2x + 6 \rightarrow x = \frac{y-6}{2} \rightarrow f^{-1}(x) = \frac{x-6}{2}$.

(31) $y = \frac{x}{3} - 4 \rightarrow x = 3(y + 4) = 3y + 12 \rightarrow f^{-1}(x) = 3x + 12$.

(35) $f \circ f^{-1}$ is the identity, so $(f \circ f^{-1})(5) = 5$.